

Indian Statistical Institute
Semester 1 (2024-2025)
M. Math. 1st Year
Final Examination
Measure Theory

Date: 29.11.24 Time: 2 hours 30 minutes

Total Points: 70

Answers must be justified with clear and precise arguments. More than one uncrossed answers to the same question or part of a question will not be entertained and only the first uncrossed answer will be graded. If you use any theorem or result proved in class state it explicitly.

1. Let f_n be a sequence of nonnegative measurable functions on $(-\infty, \infty)$ such that $f_n \rightarrow f$ a.e. m , and suppose that $\int f_n dm \rightarrow \int f dm < \infty$. Then show that for each measurable set E we have $\int_E f_n dm \rightarrow \int_E f dm$. 14 pts.
2. Let $f_n \rightarrow f$ in $L^p(\mu)$, $1 \leq p < \infty$, and let g_n be a sequence of measurable functions such that $|g_n| \leq M$ for all n , and $g_n \rightarrow g$ a.e. μ . Then show that $g_n f_n \rightarrow gf$ in $L^p(\mu)$. 14 pts.
3. (a) Let $(X, \mathcal{B}, \mu)$ be a measure space and g a nonnegative measurable function on X . Set $\nu(E) = \int_E g d\mu$. Show that ν is a measure on $\mathcal{B}$.
(b) Let f be a nonnegative measurable function on X . Then show that $\int f d\nu = \int f g d\mu$. 7 + 7 = 14 pts.
4. Use Fubini's theorem and the relation $1/x = \int_0^\infty e^{-xt} dt$, $x > 0$, to prove that $\lim_{A \rightarrow \infty} \int_0^A (\sin x/x) dx = \pi/2$. (Hint: for $(x, t) \in (0, A) \times (0, \infty)$ consider $f(x, t) = e^{-xt} \sin x$.) 14 pts.
5. Let μ, ν and λ be σ -finite measures. Show that the Radon-Nikodym derivative $d\nu/d\mu$ has the following properties: (a) If $\nu \ll \mu$ and f is a nonnegative measurable function then $\int f d\nu = \int f (d\nu/d\mu) d\mu$, (b) $d(\nu_1 + \nu_2)/d\mu = d\nu_1/d\mu + d\nu_2/d\mu$, (c) if $\nu \ll \mu \ll \lambda$, then $d\nu/d\lambda = (d\nu/d\mu)(d\mu/d\lambda)$. 4 + 4 + 6 = 14 pts.